Hüseyin Mert Çalışkan

Mechatronics Engineer | Computer Vision, 3D Graphics

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EDUCATION

Kocaeli University 2021 – Feb 2026
B.Sc. in Mechatronics Engineering Kocaeli, Türkiye

GPA: 2.71 / 4.00
Relevant coursework: Robotics, Control Systems, Image Processing, C++ Programming.
Languages: Turkish (native), English (IELTS 6.0), Japanese (JLPT N3 exam, Dec 2026)

PROJECTS

bozkır: captured ground, placed by terrain Mar 2026 – present
Independent project demo | video | code
Tools: Python (NumPy, SciPy), JavaScript, WebGL2, COLMAP, gsplat
Based on: *Gaussian Splatting Wang Tiles* (Zeng, Ma and Sander, SIGGRAPH Asia 2025)

- Built a pipeline and browser viewer that turn a few square metres of drone-photo ground, trained as 3D Gaussian splats, into tiles that cover a large generated terrain.
- Wrote a rule that reads slope, hollows and ridges, drainage and erosion sediment to place each material, and cut rivers into the terrain with water built from drone photos.
- Wrote the WebGL2 renderer with per-tile sorting in a worker and one mesh for distant ground. A world of 256 × 256 tiles (about 380 m) runs above 30 fps on a laptop GPU. About 375 automated tests.

Gaussian Splatting SLAM on an NVIDIA Jetson Orin Nano 2025
Graduation project report | code
Tools: NVIDIA Jetson Orin Nano, CUDA 12.2, Python
Based on: Gaussian-SLAM (Yugay et al.)

- Tested whether a Jetson Orin Nano can run Gaussian splatting SLAM live, mapping as the camera moves.
- Adapted Gaussian-SLAM to CUDA 12.2 and the Jetson's memory limits.

DiffCV Quest 2025
Godot game code
Tools: Godot (GDScript), Python, OpenCV

- Made a game that teaches computer vision. Its levels show edge detection and thresholding through game mechanics.

3D Asset Tools and Character Work
Indie projects
Tools: Blender (Geometry Nodes), ZBrush, Unity

- Built Blender Geometry Nodes tools: a wall generator and procedural rock and wood alpha textures.
- Took characters from high-poly sculpt in ZBrush through retopology, UV mapping and rigging in Blender.
- Modeled modular, game-ready assets with clean topology.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, C#, GLSL, C++ (basic)
- **Graphics and vision:** OpenCV, Open3D, COLMAP, 3D Gaussian splatting (gsplat), WebGL2
- **Engines and tools:** Unity, Godot, Blender, ZBrush, ROS 2, Docker, Git
- **Libraries:** NumPy, SciPy, PyTorch (basic)
- **Platforms:** Linux (Ubuntu), Windows, NVIDIA Jetson

EXPERIENCE

IT Intern, Intetra Nov 2025 – Jan 2026

Network automation and infrastructure Beykoz, Istanbul

- Ran virtual test environments in VMware for firmware recovery and network simulation.
- Configured and troubleshot L2/L3 endpoints and fixed IP conflicts with ping, ARP and static IPs.
- Wrote scripts that monitor 100+ devices (ping/HTTP) and update their firmware.

Software Developer, Autonomous Agricultural Rover Sep 2024 – Sep 2025

University team, TÜBİTAK and Teknofest supported

- Worked on SLAM-based 3D mapping with ROS 2 on a Jetson Orin.
- Integrated motor control, sensor interfacing and ROS node communication with the hardware team.

R&D Intern, Intetra Aug – Sep 2024

Traffic detection and sensor logging Gebze, Istanbul

- Built a basic vehicle detection system in Python and OpenCV for traffic videos and live IP camera streams.
- Cut the delay between stream source and host by syncing time with NTP and RTP timestamps.
- Encoded light sensor data into CRC-formatted logs for diagnostics.